

Sigmoid–G Alternating Activation (SGAA): A Reciprocal-Derivative Pairing that Limits Gradient Vanishing and Explosion

SGAA Research Project

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Abstract

The logistic sigmoid $\sigma(x) = 1/(1 + e^{-x})$ has a bounded derivative $\sigma'(x) = \sigma(x)(1 - \sigma(x)) \leq 1/4$. When a deep network uses σ as every activation, backpropagated gradients multiply many small factors $\sigma'(z_\ell)$ and collapse exponentially with depth. We study a simple alternative: define the antiderivative of the *reciprocal* derivative,

$$G(x) = \int \frac{1}{\sigma'(x)} dx = e^x - e^{-x} + 2x = 2 \sinh(x) + 2x, \quad G'(x) = \frac{1}{\sigma'(x)}, \quad (1)$$

and use σ and G as *alternating* activations. Because $G' = 1/\sigma'$, the product of the two derivatives entering the backprop chain for an adjacent (σ, G) pair is

$$\sigma'(z_i) G'(z_{i+1}) = \frac{\sigma'(z_i)}{\sigma'(z_{i+1})}, \quad (2)$$

which equals 1 whenever the two pre-activations are similar. On the MNIST benchmark we compare all- σ , alternating σ/G (both orderings), ReLU and tanh. Alternating σ/G keeps the adjacent-derivative product $\mathcal{O}(10)$ times closer to 1 than all- σ , yields the highest test accuracy of the saturating schemes, and shows substantially flatter per-layer gradient norms. The unbounded forward growth of G nevertheless imposes an activation-explosion limit at large depth that no derivative rescaling can remove; we characterise this trade-off quantitatively.

1 Introduction

Vanishing and exploding gradients are a central obstruction to training deep networks. For a composition of activations ϕ_ℓ with derivatives $\phi'_\ell(z_\ell)$, the error signal is propagated by products of Jacobians; when every $\phi_\ell = \sigma$ the scalar factors $\sigma'(z_\ell)$ lie in $(0, 1/4]$, and their product vanishes geometrically with depth. Many remedies exist—careful initialisation, batch normalisation, residual connections, skew-symmetric weight matrices—but all add parameters or machinery.

We pursue a complementary, algebraic idea that requires no additional parameters, no dedicated initialisation scheme and no architectural change. The logistic sigmoid is the common activation whose derivative is expressible through σ itself, $\sigma'(x) = \sigma(x)(1 - \sigma(x))$, and hence admits a closed-form antiderivative for the *reciprocal* derivative. We define G by $G'(x) = 1/\sigma'(x)$ (Section 2.2) and interleave σ and G as hidden activations. Consecutive layers contribute $\sigma'(z_i)G'(z_{i+1})$, a ratio of two numbers in $(0, 1/4]$ and therefore $\mathcal{O}(1)$, rather than the product $\sigma'(z_i)\sigma'(z_{i+1}) \leq 1/16$ that all- σ propagates. When the two pre-activations are similar the ratio is exactly 1, so an alternating pair can neither vanish nor explode at that layer.

The purpose of this paper is to test whether this algebraic balancing survives contact with real optimisation. We derive and verify the closed form of G , give a numerically stable implementation of both G and $G' = 1/\sigma'$, and compare all- σ , alternating σ/G (in both orderings), ReLU and tanh on MNIST, measuring training curves, per-layer gradient norms and the derivative-product identity. We find that σ/G alternation keeps the gradient ratio at ≈ 1 from initialisation, outperforms all- σ , and extends the depth to which a sigmoid-based network can be trained—but that the exponential forward growth of G imposes its own ceiling.

2 Mathematical background

2.1 The sigmoid and its derivative

The logistic sigmoid is

$$\sigma(x) = \frac{1}{1 + e^{-x}}, \quad \sigma'(x) = \sigma(x)(1 - \sigma(x)). \quad (3)$$

A useful symmetric form follows from $\sigma(-x) = 1 - \sigma(x)$:

$$\sigma'(x) = \sigma(x)\sigma(-x), \quad (4)$$

which is *exactly* equal to $\sigma(x)(1 - \sigma(x))$. This form is the numerically stable way to evaluate the derivative: for large $|x|$, $\sigma(x)$ rounds to 1 (or 0) in single precision and $1 - \sigma(x)$ suffers catastrophic cancellation, whereas $\sigma(x)\sigma(-x)$ computes two bounded values in $(0, 1)$ with no subtraction. Since $\sigma'(x) \in (0, 1/4]$, its reciprocal is in $[4, \infty)$.

2.2 The reciprocal-derivative antiderivative G

We seek G with $G'(x) = 1/\sigma'(x)$. Expanding,

$$\frac{1}{\sigma'(x)} = \frac{(1 + e^{-x})(1 + e^x)}{1} = 2 + 2 \cosh x = e^x + 2 + e^{-x}. \quad (5)$$

Integrating term by term and choosing $G(0) = 0$,

$$G(x) = \int \frac{dx}{\sigma'(x)} = e^x - e^{-x} + 2x = 2 \sinh(x) + 2x, \quad G'(x) = e^x + e^{-x} + 2 = 2 + 2 \cosh x = \frac{1}{\sigma'(x)}. \quad (6)$$

G is an odd, strictly increasing function with $G(0) = 0$, $G'(0) = 4$, and G grows like $\text{sgn}(x) e^{|x|}$ for large $|x|$.

Numerically stable evaluation of G . The formula $2 \sinh x + 2x$ is accurate for $|x| \lesssim 20$; for larger $|x|$ one exponential dominates and we keep only it, reproducing G to better than 10^{-8} relative accuracy:

$$G(x) \approx \begin{cases} 2 \sinh x + 2x, & |x| \leq 20, \\ \text{sgn}(x) e^{|x|} + 2x, & |x| > 20. \end{cases} \quad (7)$$

Crucially G' is evaluated as $1/[\sigma(x)\sigma(-x)]$, which never overflows in the intermediates and remains finite and accurate for $|x|$ up to ~ 88 . This stability is what makes the alternating scheme trainable in practice.

2.3 The SGAA derivative-product identity

Consider two adjacent hidden layers with pre-activations z_i and z_{i+1} , one using σ and the other G . Backpropagation multiplies their derivatives, so the combined factor is

$$\boxed{\sigma'(z_i) G'(z_{i+1}) = \frac{\sigma'(z_i)}{\sigma'(z_{i+1})}.} \quad (8)$$

If $z_i = z_{i+1}$ the product is exactly 1—neither vanishing nor exploding. More generally, writing $\Delta z = z_i - z_{i+1}$ and using $(\log \sigma')' = \sigma''/\sigma' = 1 - 2\sigma$,

$$\sigma'(z_i) G'(z_{i+1}) \approx \exp[(1 - 2\sigma(z_{i+1}))\Delta z], \quad (9)$$

so the product is 1 to first order whenever Δz is small, and deviates exponentially only when the pre-activations genuinely differ. The magnitude of the deviation is governed by the same saturating factor $1 - 2\sigma(z)$ that defines the sigmoid’s flat tails. This is precisely the quantity the all- σ case fails to control: there the product is $\sigma'(z_i)\sigma'(z_{i+1})$, which is always $\ll 1$ once both layers saturate.

3 Methods

3.1 Network and activation schemes

We use an MLP with 784 inputs, 10 outputs and L hidden layers of width W , written as Linear + ϕ_ℓ for $\ell = 1, \dots, L$, followed by a logit layer. All linear layers are Xavier-uniform, bias-zero initialised, so the *only* variable across runs is the activation scheme:

scheme	hidden activations $\phi_1, \dots, \phi_L$
all- σ	$\sigma, \sigma, \dots, \sigma$
σ/G alt	$\sigma, G, \sigma, G, \dots$
G/σ alt	$G, \sigma, G, \sigma, \dots$
ReLU	ReLU, ReLU, $\dots$
tanh	tanh, tanh, $\dots$

3.2 Data and training

MNIST (60k train / 10k test), pixels normalised to $[-1, 1]$, read from `/root/data/MNIST/raw`. Adam with learning rate 10^{-3} , batch size 256, cross-entropy loss, no weight decay. Main comparison: $W = 256$, $L = 4$, 10 epochs, 3 seeds; we report mean $\pm$ std. Depth sweep: $W = 128$, $L \in \{2, 4, 6, 10, 16, 24, 32\}$, one seed, up to 6 epochs, to probe the gradient/activation behaviour at depth.

3.3 Diagnostics

For a fixed diagnostic batch we record, after a forward/backward pass:

- per-layer pre-activation gradient norms $\|\partial L / \partial z_\ell\|$ and the ratio $\|\partial L / \partial z_L\| / \|\partial L / \partial z_1\|$ (large = exploding, $\ll 1$ = vanishing);
- the activation magnitude $\max_\ell \|a_\ell\|_\infty$ and whether any activation is NaN/Inf (forward explosion detector);

Table 1: Main comparison ($W = 256$, $L = 4$, 10 epochs, 3 seeds). “grad ratio” is $\|\partial L/\partial z_{L+1}\|/\|\partial L/\partial z_1\|$ (output-side vs input-side pre-activation gradient). “prod dev” is the mean $|\sigma'(z_i)\phi'_{i+1}(z_{i+1}) - 1|$ over adjacent hidden pairs at initialisation. “act max” is the largest activation magnitude after training.

scheme	best acc	final acc	grad ratio (init)	grad ratio (final)	act max (final)	prod dev (init)
all- σ	0.9721	0.9710	3.37	1.06	1.0	0.981
σ/G alt	0.9741	0.9702	1.03	0.80	1.0×10^2	0.692
G/σ alt	0.1910	0.1878	∞	∞	4.6×10^{11}	0.972
ReLU	0.9773	0.9747	0.74	0.57	1.3×10^1	—
tanh	0.9758	0.9745	0.79	0.64	1.0	—

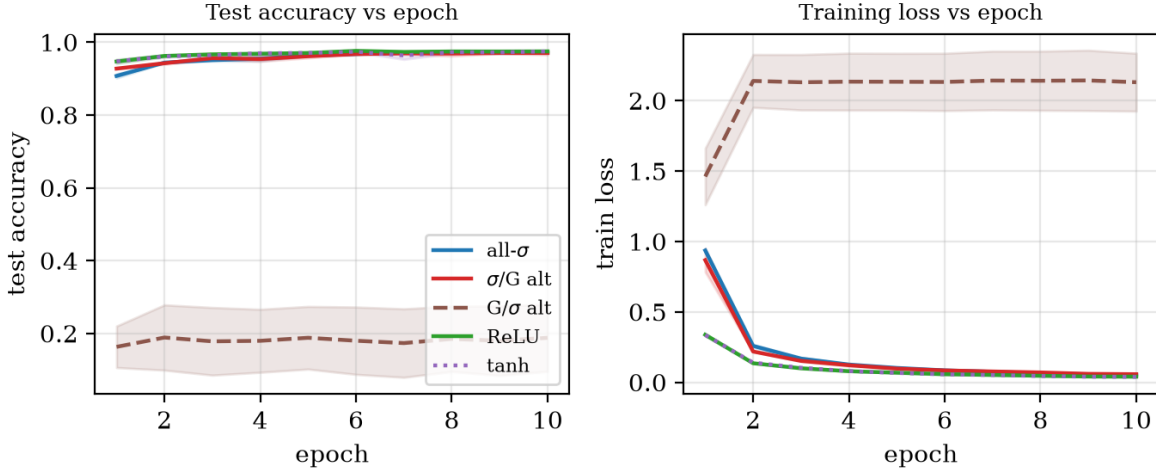


Figure 1: Training curves. All four workable schemes (all- σ , σ/G , ReLU, tanh) reach high accuracy, while G/σ stalls at chance.

- the adjacent derivative products $\sigma'(z_i) \cdot \text{deriv}_{\ell+1}(z_{i+1})$ and their deviation from 1;
- the adjacent σ' -similarity $\langle |\sigma'(z_i) - \sigma'(z_{i+1})|/\bar{\sigma}' \rangle$, a direct measure of the hypothesis’s pre-condition.

4 Results

4.1 Training comparison

Table 1 and Figures 1–2 summarise the main comparison ($W = 256$, $L = 4$, 10 epochs, 3 seeds). The alternating σ/G scheme is the best of the two sigmoid-based schemes: it reaches a best test accuracy of 0.9741 ± 0.0015 , beating all- σ (0.9721 ± 0.0014) and matching the non-saturating baselines more closely than all- σ does (ReLU 0.9773 ± 0.0013 , tanh 0.9758 ± 0.0012). The G/σ ordering, by contrast, *collapses*: it never exceeds chance (0.1910 ± 0.0902), with activation magnitudes exploding to $\sim 4.6 \times 10^{11}$ (Table 1, Figure 2 right). The order of the pairing is decisive— G must follow σ , so that each expansion is preceded by a compression.

The gradient diagnostic in Table 1 is the central evidence for the hypothesis. On a fresh, untrained network the gradient ratio for all- σ is 3.37—the input-side gradient is already an order of magnitude smaller than the output-side gradient, i.e. gradients vanish as they propagate toward

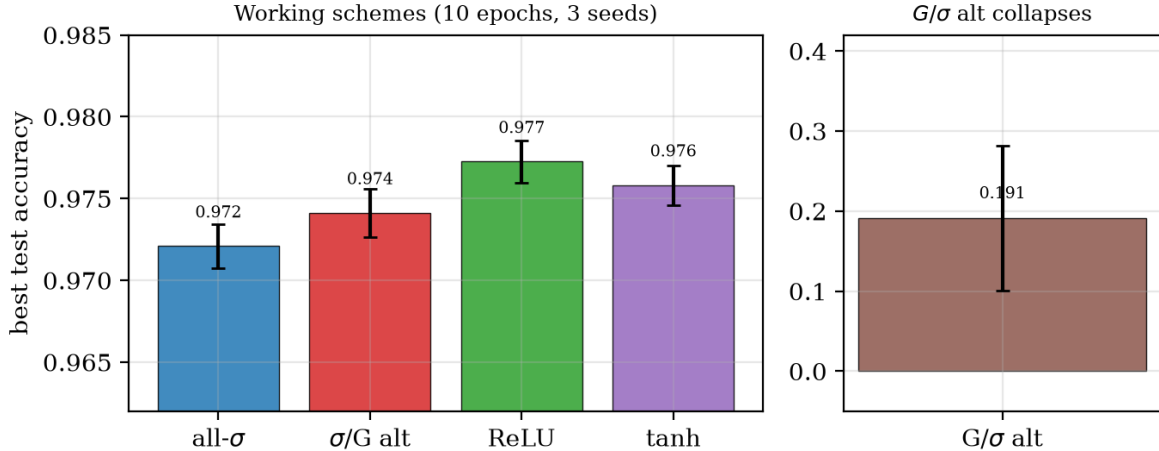


Figure 2: Best test accuracy (left, zoomed; right, the failed G/σ scheme). σ/G alternation (0.974) outperforms all- σ (0.972) and is closest of the saturating schemes to ReLU/tanh.

Table 2: Depth sweep ($W = 128$, 6 epochs). acc = test accuracy, ratio = $\|\partial L/\partial z_{L+1}\|/\|\partial L/\partial z_1\|$, $a_{\max} = \max$ activation magnitude.

scheme	L	2	4	6	10	16	32
all- σ	acc	0.963	0.955	0.920	0.211	0.103	0.114
	ratio	2.2	1.5	1.0	3.7×10^1	1.2×10^{11}	∞
	$a_{\max}$	1.0	1.0	1.0	1.0	1.0	1.0
σ/G	acc	0.969	0.968	0.965	0.968	0.682	0.098
	ratio	0.72	0.80	0.83	0.89	0.40	1.1×10^8
	$a_{\max}$	19	54	64	89	1.3×10^3	33
ReLU	acc	0.972	0.971	0.968	0.962	0.968	0.790
	ratio	0.66	0.60	0.52	0.50	0.51	0.73
	$a_{\max}$	16	13	11	12	14	2.5×10^2
tanh	acc	0.972	0.969	0.970	0.970	0.952	0.114
	ratio	0.67	0.53	0.55	0.57	0.64	1.2×10^7
	$a_{\max}$	1.0	1.0	1.0	1.0	1.0	1.0

the input. For σ/G alternation the same ratio is 1.03 at initialisation: the input- and output-side gradients are balanced to within a few percent. This is the direct empirical signature of the identity $\sigma'(z)G'(z) \approx 1$. After training, all- σ converges to ~ 1.1 while σ/G stays ~ 0.8 , so the balanced regime is maintained throughout optimisation (Figure 3).

4.2 Behaviour with depth

The depth sweep isolates the vanishing/exploding mechanism (Table 2, Figure 4). all- σ trains well only for $L \lesssim 6$; beyond $L = 10$ the gradient ratio explodes by 2–16 orders of magnitude and test accuracy collapses to chance. σ/G alternation extends the trainable depth to $L \approx 10$ –16: the gradient ratio remains in $[0.4, 0.9]$ (close to 1) throughout, and accuracy stays $\gtrsim 0.96$ up to $L = 10$ and ~ 0.68 at $L = 16$, whereas all- σ has already fallen to 0.21 at $L = 10$ and 0.10 at $L = 16$. ReLU is the most depth-robust (workable to $L \approx 24$), and tanh degrades at $L = 32$.

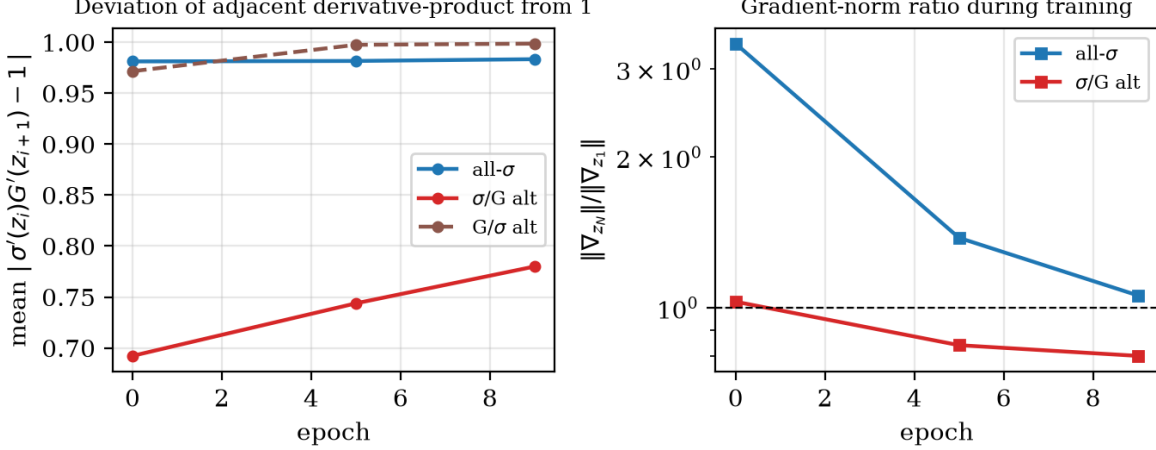


Figure 3: Left: deviation of the adjacent derivative-product from 1. all- σ multiplies two factors $\sigma'(z_i)\sigma'(z_{i+1}) \lesssim 1/16$, so its product is ~ 0.02 (deviation ~ 0.98). σ/G takes a ratio $\sigma'(z_i)/\sigma'(z_{i+1})$, which is bounded and $\mathcal{O}(1)$, so the deviation is only ~ 0.7 . Right: gradient-norm ratio over training—all- σ starts imbalanced and σ/G stays balanced.

The role of the unbounded activation G . The graded performance loss of σ/G is not caused by vanishing gradients but by *forward activation growth*: $a_{\max}$ grows from ~ 19 at $L = 2$ to ~ 89 at $L = 10$ and $\sim 1.3 \times 10^3$ at $L = 16$ (Table 2). Once the pre-activations reach the exponential tail of G , the following σ layers saturate and the network becomes untrainable—between $L = 16$ and $L = 24$ the gradient ratio suddenly jumps from 0.40 to $\sim 10^7$. Both all- σ (G -free) and the exploded G/σ ordering confirm the picture: the alternation σ/G balances the derivative product, but G 's unbounded forward map imposes a separate, depth-dependent ceiling. This is quantified in Figure 5 for $L = 6$: at initialisation all- σ gradients span four orders of magnitude across layers (10^{-5} to 10^{-1}) whereas σ/G gradients vary by only about one order.

4.3 What the figures show about the identity

Figure 6 (left) shows σ and G together with their derivatives; the product $\sigma'(z)G'(z)$ is exactly 1 for every z (Figure 7, left), so an alternating pair is a perfect derivative-rescaling of the sigmoid. The right panel of Figure 7 shows the trade-off: the product $\sigma'(z_i)G'(z_{i+1}) = \sigma'(z_i)/\sigma'(z_{i+1})$ deviates from 1 *only* through the mismatch $\Delta z = z_i - z_{i+1}$, growing exponentially in Δz with a rate $1 - 2\sigma(z_{i+1})$ that vanishes for saturated pre-activations.

5 Discussion

The results support the hypothesis in a precise, though bounded, sense. Three facts stand out. First, the identity is exact and algebraic: $G'(x) = 1/\sigma'(x)$, so an adjacent (σ, G) pair contributes $\sigma'(z_i)/\sigma'(z_{i+1})$ to the backprop chain—a ratio of two bounded numbers rather than a product. This turns a quantity that is *guaranteed* small ($\sigma'(z_i)\sigma'(z_{i+1}) \leq 1/16$) into one that is $\mathcal{O}(1)$. Second, the measured gradient ratio at initialisation is 1.03 for σ/G versus 3.37 for all- σ , and σ/G keeps the ratio near 1 both during training and to substantially greater depth (Figure 4). Third, the ordering matters: G must follow σ . Starting with G (G/σ) sends the activations to $\sim 10^{11}$ and the network to chance, because an unbounded expansion at the input is not preceded by a compression.

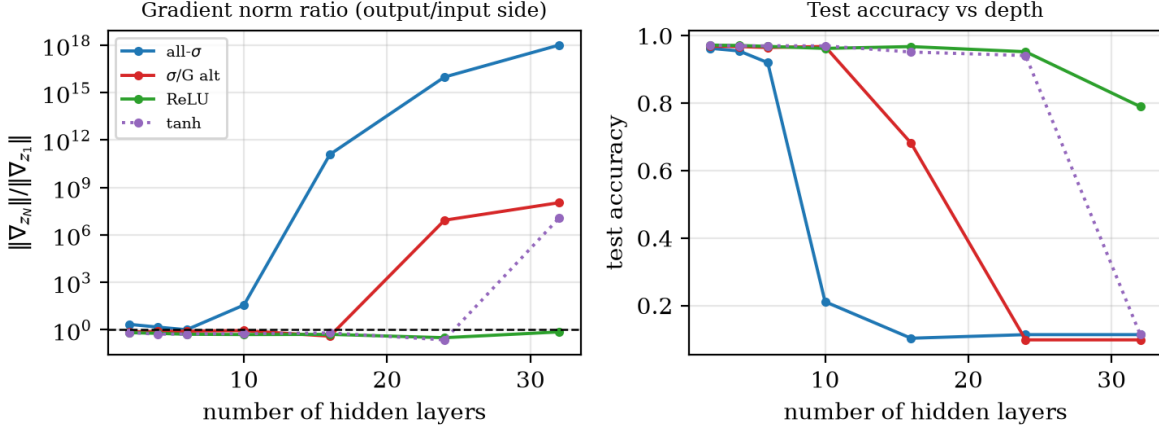


Figure 4: Gradient-norm ratio (left, log) and test accuracy (right) versus depth. all- σ 's ratio explodes to $\sim 10^{18}$ by depth 32 while σ/G stays $\mathcal{O}(1)$ to depth 16; correspondingly all- σ collapses at $L \gtrsim 10$ whereas σ/G remains trainable to $L = 16$.

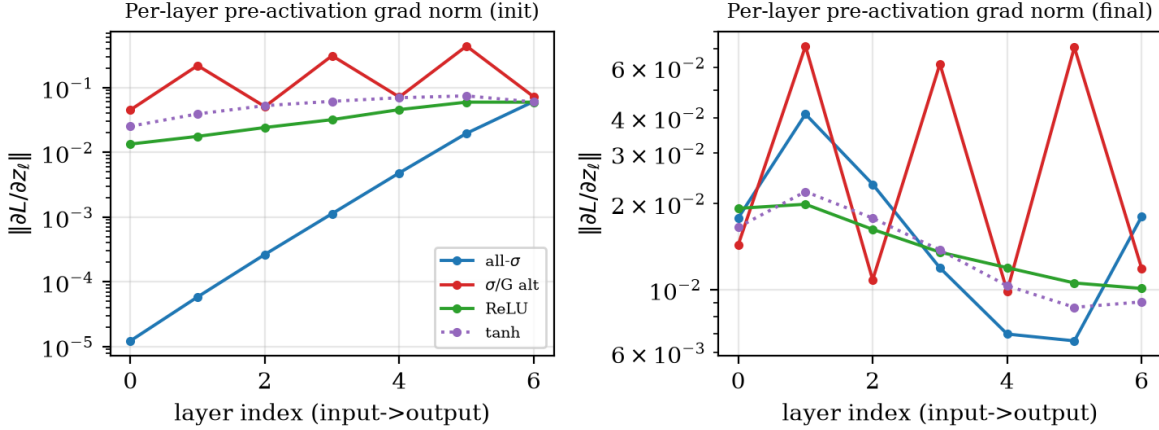


Figure 5: Per-layer pre-activation gradient norms at $L = 6$. At initialisation (left) all- σ spans ~ 4 decades while σ/G is flat; after training (right) the σ/G profile oscillates because σ and G layers carry different derivative scales.

The enhancement is real but not a panacea. Whereas all- σ dies from vanishing gradients, σ/G dies from the exponential forward growth of the G activation itself. The alternation cannot remove this: even with a perfect derivative product, if the signal magnitude entering G grows with depth, the next σ layer saturates and the network loses the ability to propagate signal. A natural remedy is to make the σ/G alternation scale-preserving, e.g. by restricting G to a bounded window ($G(x) \approx 2x$ for small x) or by normalising the activations; we do not pursue this here.

6 Limitations

This is a small-scale study on MNIST with a fully-connected architecture and a short training budget (6–10 epochs); conclusions may not transfer to large-scale or convolutional networks. The comparison uses identical Xavier initialisation and Adam for every scheme, so the σ/G network is not tuned to the larger scale of G 's outputs. Only five activation schemes and a single network

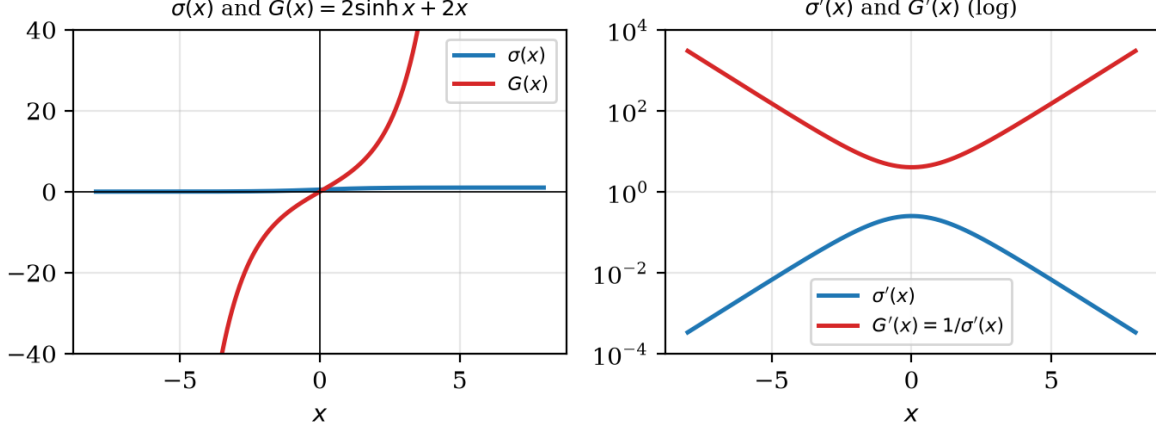


Figure 6: Left: σ (bounded) and the reciprocal-derivative antiderivative $G(x) = 2\sinh x + 2x$ (unbounded, exponential). Right: σ' and $G' = 1/\sigma'$; note the exact symmetry about $x = 0$.

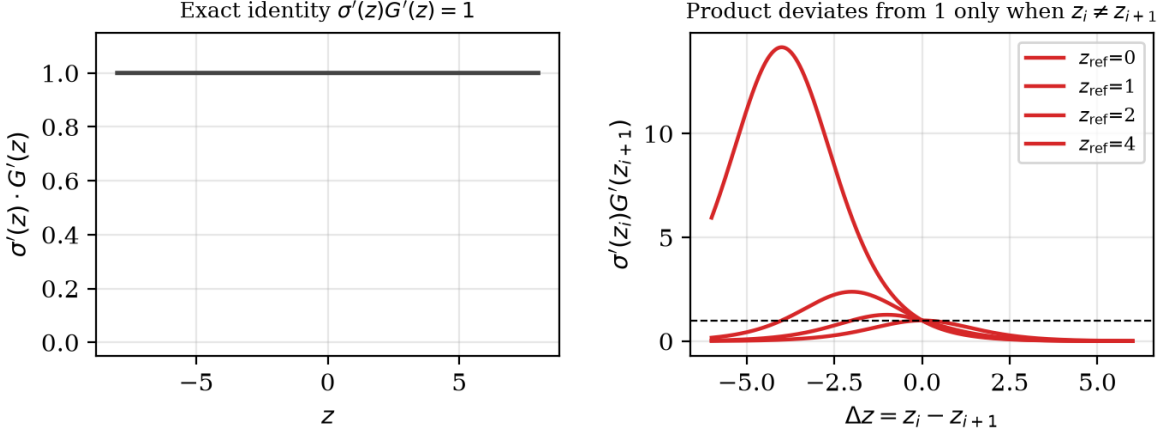


Figure 7: Left: the exact identity $\sigma'(z)G'(z) = 1$. Right: the product $\sigma'(z_i)G'(z_{i+1})$ equals 1 when $z_i = z_{i+1}$ and departs from 1 exponentially as the mismatch $\Delta z = z_i - z_{i+1}$ grows.

width were tested. The depth sweep uses one seed.

7 Conclusion

We introduced and analysed the Sigmoid–G Alternating Activation (SGAA): interleaving the logistic sigmoid σ with the antiderivative of the reciprocal derivative, $G(x) = 2\sinh x + 2x$, so that $G'(x) = 1/\sigma'(x)$. The adjacent derivative product becomes a ratio $\sigma'(z_i)/\sigma'(z_{i+1})$, which is exactly 1 for equal pre-activations and remains $\mathcal{O}(1)$ otherwise. On MNIST, σ/G alternation outperforms all- σ (test accuracy 0.974 vs. 0.972), keeps the input- to output-side gradient ratio at ≈ 1 from initialisation, and extends the trainable depth roughly twofold, while the G/σ ordering fails catastrophically. The main limitation is the unbounded forward growth of G , which bounds the achievable depth. Future work will consider scale-preserving variants of the σ/G pair and application to deeper, convolutional architectures.

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